Telangana Public Health Facility Analytics

From published geospatial layers to a clean decision dashboard

Project type	Data analytics and Power BI preparation
Geography	Telangana, India
Source	TGRAC Health Facilities Mapping service
Prepared by	Harika

This report documents the source, preparation logic, dashboard design, findings, limitations, and reproducibility checks.

1. Executive summary

Public health facility locations are available through separate map layers. The project combines those layers into one analysis table so that counts, geographic coverage, facility type mix, and record completeness can be examined consistently.

Clean records	**2,628**
Districts represented	**33**
Coordinates present	**100.0%**
Exact duplicates removed	**20**
Facility categories	**12**

Main outcome

The result is a traceable CSV, a quality summary, a Power BI measure set, and a dashboard image. The analysis deliberately separates a published record count from service capacity. A facility appearing in the dataset does not confirm its current staffing, beds, medicines, or operating status.

Why this is useful

A consistent table allows analysts to compare geographic representation without repeatedly joining source layers. The completeness measure also shows where a map is ready for location analysis and where contextual fields still need review.

2. Problem and analytical questions

The source is optimised for mapping rather than tabular analysis. Each facility group has its own endpoint, fields can be null, and the same place can appear more than once. The project therefore begins with a data engineering problem before any chart is created.

Questions answered

How many unique published facilities remain after exact deduplication? Which facility types contribute most records? Which districts have the largest record counts? How complete are the location and administrative fields? Can every processed row be traced back to a source layer?

Root cause

The most important root cause is structural fragmentation. Separate layers make category totals easy to view but difficult to reproduce as a single governed dataset. Null values and spelling variations create a second source of inconsistency.

Success criteria

Criterion	Test
Traceability	Every row has a source layer and project identifier
Consistency	Text is trimmed and district variants are standardised
Uniqueness	Exact duplicate facility signatures are removed
Usability	Output loads directly into Power BI
Honesty	Dashboard states what the data cannot prove

3. Data source and provenance

The data comes from the Telangana Remote Sensing Applications Centre Health Facilities Mapping ArcGIS service. Twelve public layers are queried through the ArcGIS REST interface. The raw JSON responses and a retrieval manifest are stored before transformation.

Source fields used

Source concept	Prepared field
OBJECTID and layer identifier	facility_id
Facility_Name	facility_name
Facility_Type or layer label	facility_type
District and District_1	district
Latitude or geometry y	latitude
Longitude or geometry x	longitude

Provenance safeguards

Raw responses are not overwritten. The manifest records the publisher, endpoint, retrieval time, and requested layer identifiers. The processing script can be rerun when the source changes, allowing a later result to be compared with this snapshot.

Interpretation boundary

This project describes the records published by the service. It is not an official census of all facilities and does not establish availability, quality, or clinical performance.

4. Preparation pipeline

The pipeline follows five stages: collect, preserve, standardise, validate, and publish. Each API response is saved before selected fields are converted into the project schema.

Stage	Method	Output
Collect	Query each facility layer	Raw JSON
Preserve	Write response and source manifest	Traceable snapshot
Standardise	Clean spacing, case, and district variants	Consistent columns
Validate	Remove exact signatures and score completeness	Quality metrics
Publish	Sort and export a flat table	Power BI ready CSV

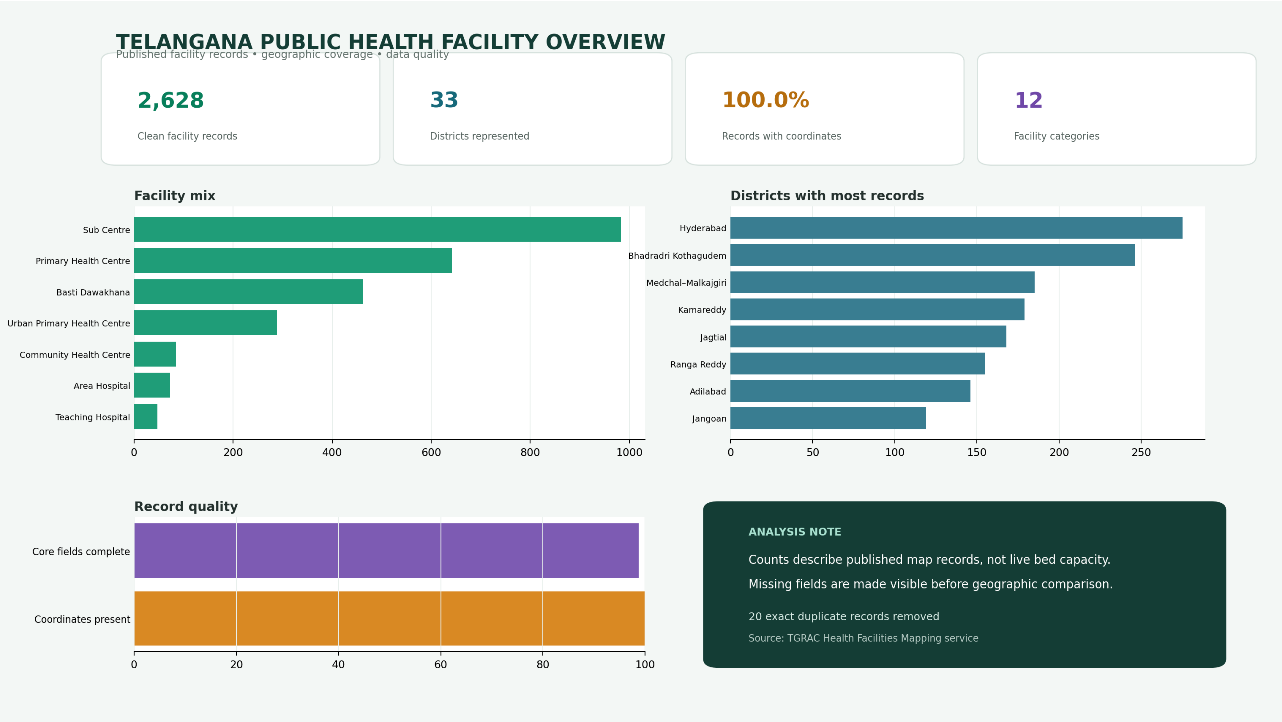
Deduplication rule

An exact signature uses the case-insensitive facility name, district, latitude, and longitude. This conservative rule avoids deleting similarly named facilities in different places. Near duplicates remain visible for manual investigation.

Completeness rule

The score checks name, type, district, mandal, latitude, and longitude. It measures whether important analysis fields are present; it does not measure whether each value is factually correct.

5. Dashboard design



The dashboard begins with four KPIs and then moves from facility mix to district concentration and record quality. The layout keeps the main caution next to the charts so the published counts are not mistaken for current capacity.

6. Facility mix findings

The published source is weighted toward primary and community-level records. This makes facility type an essential filter when interpreting district totals.

Facility type	Records	Share
Sub Centre	983	37.4%
Primary Health Centre	642	24.4%
Basti Dawakhana	462	17.6%
Urban Primary Health Centre	289	11.0%
Community Health Centre	85	3.2%
Area Hospital	73	2.8%

Interpretation

A district with many sub centres may have a high overall facility count without having many hospital records. Comparisons should therefore use both total records and type-specific records. The dashboard is designed to support that drill down in Power BI.

Actionable next step

Add population denominators and road travel times only after matching geographic definitions and dates. A facility per population measure would otherwise combine incompatible snapshots.

7. Geographic and quality findings

The cleaned table represents 33 district labels. Hyderabad has the largest number of published records in this snapshot, followed by Bhadradri Kothagudem. This ranking describes source coverage and facility mix rather than need or performance.

District	Records
Hyderabad	275
Bhadradri Kothagudem	246
Medchal–Malkajgiri	185
Kamareddy	179
Jagtial	168
Ranga Reddy	155
Adilabad	146
Jangoan	119

All prepared rows include coordinates because the service geometry is used when explicit latitude or longitude fields are empty. Most rows also meet the core completeness threshold. The remaining gaps occur mainly in administrative context such as mandal.

8. Power BI model and measures

The clean table can be loaded as a single facility dimension for this scope. Latitude and longitude are decimal numbers; district, facility type, department, and source layer are categorical fields. The project includes reusable DAX measures for totals, district coverage, mapping coverage, and average completeness.

Visual	Field or measure	Purpose
KPI card	Total Facilities	Scale of the source snapshot
KPI card	District Count	Geographic representation
Bar chart	Facility type and Total Facilities	Mix of record categories
Bar chart	District and Total Facilities	Concentration of published records
Map	Latitude, longitude, facility name	Spatial exploration
Quality card	Average Completeness	Preparation risk visibility

Recommended interaction

Selecting a facility type should filter every district visual and the map. Selecting a district should show the facility list and completeness details. Tooltips should repeat the source layer and avoid unverified service claims.

9. Validation, limitations, and next steps

Automated tests check text cleaning, district normalisation, coordinate fallback, and exact duplicate removal. The continuous integration workflow runs tests and static checks for every update.

Limitations

The source can change after retrieval. A record can be outdated even when its fields are complete. The analysis has no population denominator, operating status, service hours, staffing, beds, medicine inventory, or quality indicators. Coordinate presence is not independently geocoded.

Next steps

Compare retrieval snapshots to detect additions and removals. Review near duplicate names. Add verified district population data with matching boundaries. Introduce travel time analysis for a carefully chosen service type. Request publisher confirmation before using the output operationally.

Reproducibility

Create a Python environment, install the pinned requirements, run the pipeline, render the dashboard, build this report, and execute the test suite. The README contains the exact commands and the manifest records the source endpoint.